

Multi-Agent Pac-Man

CSE 537 Project 2 - Questions 1-3 and performance analysis

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Team information for this submission.

1. Method and measurement

All measurements use the included starter layouts and Python 3.13.15. Each decision starts from a new initial GameState. Runtime is the median of 7 untraced wall-clock runs; peak Python allocation is the median of 5 separate tracemalloc runs. Layout loading, graphics, and complete games are excluded from decision measurements. Expanded states count nonterminal legal-action queries; generated successors count calls to generateSuccessor. ReflexAgent is reported by its immediate successors because it has no search tree. Complete-game trials use the fixed random seed cs188.

2. Question 1: ReflexAgent

For each legal action, the agent evaluates the immediate successor state. It starts with the game score, adds $12/(\text{distance to nearest food} + 1)$, subtracts 4 per remaining food and 6 per remaining capsule, and adds 20 when the move eats food. Stopping costs 15. A live ghost at distance 1 costs 500 and at distance 2 costs 80; a collision is rejected. A scared ghost earns up to $30/(\text{distance} + 1)$ when its timer leaves enough time to reach it. These terms reward food progress while reacting to nearby danger and temporary capture opportunities.

Layout	Ghosts	Action	Successors	Time (ms)	Peak (KiB)
testClassic	1	East	3	0.050	2.73
openClassic	1	West	4	0.148	10.54
mediumClassic	1	East	3	0.126	13.97
mediumClassic	2	East	3	0.136	14.15

Table 1. ReflexAgent initial-decision measurements.

Layout	Ghosts	Wins	Average score
testClassic	1	10/10	540.9
openClassic	1	10/10	1233.1
mediumClassic	1	10/10	1467.2
mediumClassic	2	9/10	1711.8

Table 2. Ten fixed-seed games per layout. The openClassic target allows no more than two losses.

The agent won all ten games on testClassic and openClassic. With two ghosts on mediumClassic it won nine of ten. The remaining loss shows the limit of a one-step decision: it does not model a future ghost move.

3. Question 2: MinimaxAgent

Pac-Man (agent 0) maximizes value; every ghost minimizes it. Agents move in index order, and depth increases only after the last ghost responds. A depth of one therefore contains one Pac-Man move and all ghost moves. Recursion stops at the requested depth, a win, a loss, or a state with no legal actions. Leaves use the configured `self.evaluationFunction`, which defaults to the game score. Each successor is generated once.

Depth	Action	Value	Expanded	Generated	Time (ms)	Peak (KiB)
1	West	9	22	33	0.774	7.65
2	West	8	155	211	2.211	12.96
3	Stop	7	726	1160	13.348	19.45
4	West	-492	4178	5916	60.341	24.05

Table 3. Minimax initial decisions on `minimaxClassic`. Values match the assignment's 9, 8, 7, and -492 reference sequence.

The depth-4 search generates 5,916 successors, compared with 33 at depth 1. The negative depth-4 value is consistent with worst-case ghost choices; random ghosts in a played game need not choose those responses.

4. Question 3: AlphaBetaAgent

Alpha-beta follows the same state, turn, depth, and leaf rules. Pac-Man updates alpha and ghosts update beta. A branch is cut only when its value is strictly beyond the opposite bound, so equal-valued branches are still explored. Actions remain in `getLegalActions` order, as the assignment requires for expansion-count grading.

Depth	Action	Value	Expanded	Generated	Time (ms)	Peak (KiB)
1	West	9	14	19	0.211	7.70
2	West	8	136	179	1.900	12.98
3	Stop	7	592	886	9.391	19.49
4	West	-492	3239	4463	47.148	24.07

Table 4. Alpha-beta preserves every `minimaxClassic` value while visiting fewer states.

5. Search-efficiency comparison

Depth	Expanded: M / AB	Reduction	Generated: M / AB	Reduction	Time: M / AB (ms)
1	22 / 14	36.4%	33 / 19	42.4%	0.774 / 0.211
2	155 / 136	12.3%	211 / 179	15.2%	2.211 / 1.900
3	726 / 592	18.5%	1160 / 886	23.6%	13.348 / 9.391
4	4178 / 3239	22.5%	5916 / 4463	24.6%	60.341 / 47.148

Table 5. M = Minimax, AB = Alpha-beta. All decisions begin from the same minimaxClassic state.

Strategy	Value	Expanded	Generated	Time (ms)	Peak (KiB)
Minimax	27	397	828	13.466	22.25
Alpha-beta	27	143	211	3.542	22.21

Table 6. smallClassic at depth 3; both strategies choose West.

Interpretation

At depth 4, alpha-beta expanded 22.5% fewer states and generated 24.6% fewer successors while returning the same -492 value. On smallClassic at depth 3, it generated 74.5% fewer successors. Pruning benefits depend on the supplied action order; this implementation does not reorder children. Both algorithms use depth-first recursion, so their peak traced memory is similar despite different work counts.

Limits and verification

Decision tables describe one initial-state choice, not an entire game. Later states can have different branching factors. Tracemalloc reports Python allocations during the decision, not whole-process memory. Absolute times depend on the machine and Python version. Five local regression tests cover multi-ghost turns, reference values, reference expansion counts, and strict equality behavior in alpha-beta. Headless game tests cover the specified layouts. No official autograder was included in the starter package.

Conclusion

Question 1 shows that a low-cost, one-step evaluation can clear the required layouts when food progress and ghost danger are both represented. Question 2 supplies correct multi-ghost adversarial values at increasing cost. Question 3 keeps those values and reduces the number of visited states through pruning, especially on deeper searches.